Class 12 Mathematics

Chapter 4: Determinants

Practice Set A: step-wise solutions

20 questions **54** marks **3** levels

Each row is one line of working. The red number in the right margin is what an examiner gives for that line, as in the CBSE marking scheme (½ is half a mark). Write every line that carries a mark. ✓ ½

Level 1

8 marks

1 **MCQ** PYQ-style — 1 mark

If $\begin{vmatrix} x+1 & x-1 \\ x-3 & x+2 \end{vmatrix} = \begin{vmatrix} 4 & -1 \\ 1 & 3 \end{vmatrix}$, then x is equal to:

(A) $\frac{12}{7}$ **(B)** 2 **(C)** -2 **(D)** $\frac{8}{7}$

Formula / idea: $\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$

1 Expand the LHS.
$(x+1)(x+2) - (x-1)(x-3) = (x^2+3x+2) - (x^2-4x+3) = 7x - 1$

2 Expand the RHS.
$4(3) - (-1)(1) = 12 + 1 = 13$

3 Equate and solve. ✓ 1
$7x - 1 = 13 \Rightarrow x = 2$

4 Hence option (B).

Answer **(B)** 2

Watch out: Writing the RHS as $12 - 1 = 11$; here $bc = (-1)(1) = -1$, so $ad - bc = 12 + 1$.

2 **MCQ** CBSE 2023 — 1 mark

Let A be a 3×3 matrix such that $|\text{adj } A| = 64$. Then $|A|$ is equal to:

(A) 8 only **(B)** -8 only **(C)** 64 **(D)** 8 **or** -8

Formula / idea: For a square matrix of order n: $|\text{adj } A| = |A|^{n-1}$.

1 Use the result with $n = 3$.
$|\text{adj } A| = |A|^{3-1} = |A|^2$

2 Put $|\text{adj } A| = 64$ and solve. ✓ 1
$|A|^2 = 64 \Rightarrow |A| = \pm 8$

3 Both values are possible. Option (D).

Answer **(D)** 8 **or** -8

Watch out: Taking only the positive square root; $|A|$ can be negative.

3 MCQ CBSE 2023 — 1 mark

If $|A| = 2$, where A is a 2×2 matrix, then $|4A^{-1}|$ equals:

(A) 4 **(B)** 2 **(C)** 8 **(D)** $\frac{1}{32}$

Formula / idea: $|kB| = k^n|B|$ for order n, and $|A^{-1}| = \frac{1}{|A|}$.

1 Find $|A^{-1}|$.
$|A^{-1}| = \frac{1}{|A|} = \frac{1}{2}$

2 Use $|kB| = k^n|B|$ with $k = 4$, $n = 2$. ✓ 1
$|4A^{-1}| = 4^2 \cdot \frac{1}{2} = 16 \cdot \frac{1}{2} = 8$

3 Hence option (C).

Answer **(C)** 8

Watch out: Writing $|4A^{-1}| = 4 \cdot \frac{1}{2} = 2$; the factor 4 comes out as 4^2 for a 2×2 matrix.

4 MCQ CBSE SQP 2024-25 — 1 mark

If the points (x_1, y_1), (x_2, y_2) and $(x_1 + x_2,\ y_1 + y_2)$ are collinear, then x_1y_2 is equal to:

(A) x_2y_1 **(B)** x_1y_1 **(C)** x_2y_2 **(D)** x_1x_2

Formula / idea: Three points are collinear when the area of the triangle they form is 0.

1 Collinear, so the area determinant is zero.
$$\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_1 + x_2 & y_1 + y_2 & 1 \end{vmatrix} = 0$$

2 Expand along R_1.
$x_1(y_2 - y_1 - y_2) - y_1(x_2 - x_1 - x_2) + (x_2y_1 + x_2y_2 - x_1y_2 - x_2y_2) = 0$

3 Simplify.
$-x_1y_1 + x_1y_1 + x_2y_1 - x_1y_2 = 0 \Rightarrow x_1y_2 = x_2y_1$ ✓ 1

4 Hence option (A).

Answer **(A)** x_2y_1

5 MCQ PYQ-style 1 mark

If A_{ij} is the cofactor of the element a_{ij} of the determinant $\begin{vmatrix} 5 & 3 & 8 \\ 2 & 0 & 1 \\ 1 & 2 & 3 \end{vmatrix}$, then the value of $a_{32} \cdot A_{32}$ is:

(A) 22 **(B)** -22 **(C)** 11 **(D)** -11

Formula / idea: $A_{ij} = (-1)^{i+j} M_{ij}$, where M_{ij} is the minor of a_{ij}.

1 Read the element in row 3, column 2.
$a_{32} = 2$

2 Minor: delete row 3 and column 2.
$M_{32} = \begin{vmatrix} 5 & 8 \\ 2 & 1 \end{vmatrix} = 5 - 16 = -11$

3 Cofactor: attach the sign $(-1)^{3+2}$.
$A_{32} = (-1)^5(-11) = 11$

4 Multiply.
$a_{32} \cdot A_{32} = 2 \times 11 = 22$ ✓ 1

5 Hence option (A).

Answer **(A)** 22

Watch out: Using the minor -11 in place of the cofactor and getting -22.

6 **Assertion-Reason** Board-pattern 1 mark

Assertion (A): If A is a square matrix of order 3 and $|A| = 3$, then $(\text{adj } A)\, A = 3I$. **Reason (R):** For a square matrix A of order n, $|\text{adj } A| = |A|^{n-1}$.

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

(B) **Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).**

(C) Assertion (A) is true, but Reason (R) is false.

(D) Assertion (A) is false, but Reason (R) is true.

Formula / idea: $A(\text{adj } A) = (\text{adj } A)A = |A|\, I$ for every square matrix A.

1 Check A using $(\text{adj } A)A = |A|\, I$ with $|A| = 3$.
$(\text{adj } A)\, A = 3I$

2 So **A is true.**

3 R is the standard result for order n. **R is true.**

4 R gives the value of $|\text{adj } A|$, not the product $(\text{adj } A)A$. So R does not explain A.

5 Both true, R not the explanation. Hence option (B). ✓ 1

Answer **(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).**

Watch out: Choosing (A) because both involve $\text{adj } A$; A follows from $(\text{adj } A)A = |A|\, I$, not from R.

7 **Short answer** PYQ-style 2 marks

Find the values of k if the area of the triangle with vertices $(2, -6)$, $(5, 4)$ and $(k, 4)$ is 35 sq units.

Formula / idea: Area $= \frac{1}{2}\left|\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}\right|$

1 Write the area using determinants. ✓ ½

$$\Delta = \frac{1}{2}\begin{vmatrix} 2 & -6 & 1 \\ 5 & 4 & 1 \\ k & 4 & 1 \end{vmatrix} = \pm 35$$

The determinant may be positive or negative.

2 Expand along R_1. ✓ ½

$$\frac{1}{2}\left[2(4-4) + 6(5-k) + 1(20-4k)\right] = \frac{1}{2}(50 - 10k) = 25 - 5k$$

3 Equate to ± 35.
$25 - 5k = 35 \quad \text{or} \quad 25 - 5k = -35$ ✓ ½

4 Solve each equation.
$k = -2 \quad \text{or} \quad k = 12$ ✓ ½

5 Hence $k = 12$ or $k = -2$.

Answer $k = 12$ **or** $k = -2$

Watch out: Taking only $+35$ and missing the second value $k = 12$.

Level 2

17 marks

8 **Short answer** PYQ-style 2 marks

Find the adjoint and the inverse of the matrix $A = \begin{bmatrix} -1 & 5 \\ -3 & 2 \end{bmatrix}$.

Formula / idea: $A^{-1} = \dfrac{1}{|A|} \operatorname{adj} A$, when $|A| \neq 0$.

1 Find $|A|$.
$|A| = (-1)(2) - (5)(-3) = -2 + 15 = 13 \neq 0$
So A^{-1} exists. ✓ ½

2 For a 2×2 matrix: swap the diagonal, change signs of the other two.
$\operatorname{adj} A = \begin{bmatrix} 2 & -5 \\ 3 & -1 \end{bmatrix}$ ✓ 1

3 Divide by $|A|$.
$A^{-1} = \dfrac{1}{|A|} \operatorname{adj} A = \dfrac{1}{13} \begin{bmatrix} 2 & -5 \\ 3 & -1 \end{bmatrix}$ ✓ ½

Answer $\operatorname{adj} A = \begin{bmatrix} 2 & -5 \\ 3 & -1 \end{bmatrix}$, $A^{-1} = \dfrac{1}{13} \begin{bmatrix} 2 & -5 \\ 3 & -1 \end{bmatrix}$

Watch out: Changing the signs of the diagonal elements instead of the off-diagonal ones.

9 **Short answer** PYQ-style 3 marks

For the matrix $A = \begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix}$, find the numbers a and b such that $A^2 + aA + bI = O$. Hence find A^{-1}.

1 Find A^2.
$A^2 = \begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 11 & 8 \\ 4 & 3 \end{bmatrix}$ ✓ 1

2 Substitute in $A^2 + aA + bI = O$.
$\begin{bmatrix} 11+3a+b & 8+2a \\ 4+a & 3+a+b \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$ ✓ ½

3 Equate the $(2,1)$ elements, then the $(1,1)$ elements.
$4 + a = 0 \Rightarrow a = -4; \qquad 11 - 12 + b = 0 \Rightarrow b = 1$ ✓ ½

4 So $A^2 - 4A + I = O$. Check that A^{-1} exists.
$|A| = 3 - 2 = 1 \neq 0$

5 Multiply both sides by A^{-1}.
$A - 4I + A^{-1} = O \;\Rightarrow\; A^{-1} = 4I - A$ ✓ ½

6 Compute $4I - A$.
$A^{-1} = \begin{bmatrix} 4 & 0 \\ 0 & 4 \end{bmatrix} - \begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -2 \\ -1 & 3 \end{bmatrix}$ ✓ ½

Answer $a = -4, b = 1$; $A^{-1} = \begin{bmatrix} 1 & -2 \\ -1 & 3 \end{bmatrix}$

Watch out: Writing $A^{-1}I$ as I while multiplying through; it is A^{-1}.

10 **Short answer** PYQ-style 2 marks

Using determinants, find the equation of the line joining $A(1,3)$ and $B(0,0)$. Hence find k if $D(k,0)$ is a point such that the area of $\triangle ABD$ is 3 sq units.

Formula / idea: $P(x,y)$ lies on line AB when the area of $\triangle PAB$ is 0.

1 Let $P(x,y)$ be any point on AB. Area of $\triangle PAB = 0$.
$\begin{vmatrix} x & y & 1 \\ 1 & 3 & 1 \\ 0 & 0 & 1 \end{vmatrix} = 0$

2 Expand along R_1.
$x(3-0) - y(1-0) + 1(0-0) = 0 \;\Rightarrow\; y = 3x$ ✓ 1

3 Area of $\triangle ABD$ using determinants.
$\frac{1}{2}\begin{vmatrix} 1 & 3 & 1 \\ 0 & 0 & 1 \\ k & 0 & 1 \end{vmatrix} = \pm 3$

4 Expand along R_1.
$\frac{1}{2}\left[1(0-0) - 3(0-k) + 1(0-0)\right] = \frac{3k}{2} = \pm 3$ ✓ ½

5 Solve.

$k = \pm 2$

✓ ½

Answer $y = 3x$; $k = \pm 2$

Watch out: Forgetting the $\pm$ sign for the area and giving only $k = 2$.

11 **Short answer** PYQ-style 3 marks

Solve the following system of equations by the matrix method: $2x + 5y = 1$, $3x + 2y = 7$.

Formula / idea: $AX = B \Rightarrow X = A^{-1}B$, when $|A| \neq 0$.

1 Write the system as $AX = B$.

$$\begin{bmatrix} 2 & 5 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 \\ 7 \end{bmatrix}$$

✓ ½

2 Find $|A|$.

$|A| = 4 - 15 = -11 \neq 0$

So a unique solution exists.

✓ ½

3 Find $\text{adj}\, A$.

$$\text{adj}\, A = \begin{bmatrix} 2 & -5 \\ -3 & 2 \end{bmatrix}$$

✓ ½

4 Find A^{-1}.

$$A^{-1} = -\frac{1}{11}\begin{bmatrix} 2 & -5 \\ -3 & 2 \end{bmatrix}$$

✓ ½

5 Find $X = A^{-1}B$.

$$X = -\frac{1}{11}\begin{bmatrix} 2 - 35 \\ -3 + 14 \end{bmatrix} = -\frac{1}{11}\begin{bmatrix} -33 \\ 11 \end{bmatrix} = \begin{bmatrix} 3 \\ -1 \end{bmatrix}$$

✓ ½

6 Hence $x = 3$, $y = -1$.

✓ ½

Answer $x = 3$, $y = -1$

Watch out: Writing $X = BA^{-1}$; the order must be $A^{-1}B$.

12 **Short answer** Board-pattern 2 marks

Examine the consistency of the system of equations: $2x - 3y = 5,\ 6x - 9y = 15$.

Formula / idea: If $|A| = 0$ and $(\text{adj } A)B = O$, the system may be consistent or inconsistent; check the equations.

1 Write as $AX = B$ and find $|A|$.
$A = \begin{bmatrix} 2 & -3 \\ 6 & -9 \end{bmatrix},\ B = \begin{bmatrix} 5 \\ 15 \end{bmatrix},\ |A| = -18 + 18 = 0$ ✓ ½

2 Find $\text{adj } A$.
$\text{adj } A = \begin{bmatrix} -9 & 3 \\ -6 & 2 \end{bmatrix}$ ✓ ½

3 Find $(\text{adj } A)B$.
$(\text{adj } A)B = \begin{bmatrix} -45 + 45 \\ -30 + 30 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} = O$ ✓ ½

4 The second equation is 3 times the first, so both are the same line.

5 Hence the system is consistent, with infinitely many solutions. ✓ ½

Answer **Consistent, with infinitely many solutions.**

Watch out: Calling the system inconsistent just because $|A| = 0$.

13 **Short answer** Board-pattern 2 marks

If $A = \begin{bmatrix} 3 & -4 \\ 2 & -3 \end{bmatrix}$, show that $A^{-1} = A$.

1 Find $|A|$.
$|A| = 3(-3) - (-4)(2) = -9 + 8 = -1 \neq 0$ ✓ ½

2 Find $\text{adj } A$.
$\text{adj } A = \begin{bmatrix} -3 & 4 \\ -2 & 3 \end{bmatrix}$ ✓ ½

3 Use $A^{-1} = \dfrac{1}{|A|}\,\text{adj } A$.
$A^{-1} = \dfrac{1}{-1}\begin{bmatrix} -3 & 4 \\ -2 & 3 \end{bmatrix} = \begin{bmatrix} 3 & -4 \\ 2 & -3 \end{bmatrix}$ ✓ ½

4 This is the matrix A. Hence $A^{-1} = A$. ✓ ½

Answer $A^{-1} = \begin{bmatrix} 3 & -4 \\ 2 & -3 \end{bmatrix} = A$

Watch out: Dropping the minus sign of $|A| = -1$ when dividing.

14 **Short answer** PYQ-style 3 marks

If $A = \begin{bmatrix} 3 & 7 \\ 2 & 5 \end{bmatrix}$ and $B = \begin{bmatrix} 6 & 8 \\ 7 & 9 \end{bmatrix}$, verify that $(AB)^{-1} = B^{-1}A^{-1}$.

1 Find AB. ✓ ½

$$AB = \begin{bmatrix} 18+49 & 24+63 \\ 12+35 & 16+45 \end{bmatrix} = \begin{bmatrix} 67 & 87 \\ 47 & 61 \end{bmatrix}$$

2 Find $|AB|$.

$$|AB| = 67 \times 61 - 87 \times 47 = 4087 - 4089 = -2$$

3 Find $(AB)^{-1}$. ✓ 1

$$(AB)^{-1} = -\frac{1}{2}\begin{bmatrix} 61 & -87 \\ -47 & 67 \end{bmatrix}$$

4 Find A^{-1}; here $|A| = 15 - 14 = 1$. ✓ ½

$$A^{-1} = \begin{bmatrix} 5 & -7 \\ -2 & 3 \end{bmatrix}$$

5 Find B^{-1}; here $|B| = 54 - 56 = -2$. ✓ ½

$$B^{-1} = -\frac{1}{2}\begin{bmatrix} 9 & -8 \\ -7 & 6 \end{bmatrix}$$

6 Find $B^{-1}A^{-1}$.

$$B^{-1}A^{-1} = -\frac{1}{2}\begin{bmatrix} 45+16 & -63-24 \\ -35-12 & 49+18 \end{bmatrix} = -\frac{1}{2}\begin{bmatrix} 61 & -87 \\ -47 & 67 \end{bmatrix}$$

7 Both sides are equal. Hence $(AB)^{-1} = B^{-1}A^{-1}$ is verified. ✓ ½

Answer $(AB)^{-1} = B^{-1}A^{-1} = -\frac{1}{2}\begin{bmatrix} 61 & -87 \\ -47 & 67 \end{bmatrix}$. **Verified.**

Watch out: Multiplying in the order $A^{-1}B^{-1}$; the reversal law needs $B^{-1}A^{-1}$.

Level 3

29 marks

15 **Long answer** CBSE SQP 2024-25 5 marks

The equation of the path traversed by the ball headed by a footballer is $y = ax^2 + bx + c$ (where $0 \le x \le 14$ and $a, b, c \in \mathbb{R}$, $a \neq 0$) with respect to an XY-coordinate system in the vertical plane. The ball passes through the points $(2, 15)$, $(4, 25)$ and $(14, 15)$. Determine the values of a, b and c by solving the system of linear equations in a, b and c, using the matrix method. Also find the equation of the path traversed by the ball.

Formula / idea: $AX = B \Rightarrow X = A^{-1}B = \dfrac{1}{|A|}(\text{adj } A)B.$

1 Put each point in $y = ax^2 + bx + c$. ✓ 1

$4a + 2b + c = 15, \quad 16a + 4b + c = 25, \quad 196a + 14b + c = 15$

2 Write as $AX = B$.

$$\begin{bmatrix} 4 & 2 & 1 \\ 16 & 4 & 1 \\ 196 & 14 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 15 \\ 25 \\ 15 \end{bmatrix}$$

3 Find $|A|$ by expanding along R_1. ✓ ½

$|A| = 4(4 - 14) - 2(16 - 196) + 1(224 - 784) = -40 + 360 - 560 = -240 \neq 0$

4 Cofactors of row 1. ✓ ½

$A_{11} = -10, \quad A_{12} = 180, \quad A_{13} = -560$

5 Cofactors of row 2. ✓ ½

$A_{21} = 12, \quad A_{22} = -192, \quad A_{23} = 336$

6 Cofactors of row 3. ✓ ½

$A_{31} = -2, \quad A_{32} = 12, \quad A_{33} = -16$

7 $\text{adj } A$ is the transpose of the cofactor matrix. ✓ ½

$$\text{adj } A = \begin{bmatrix} -10 & 12 & -2 \\ 180 & -192 & 12 \\ -560 & 336 & -16 \end{bmatrix}$$

8 Write A^{-1}.

$$A^{-1} = -\frac{1}{240}\begin{bmatrix} -10 & 12 & -2 \\ 180 & -192 & 12 \\ -560 & 336 & -16 \end{bmatrix}$$

9 Find $X = A^{-1}B$. ✓ ½

$$X = -\frac{1}{240}\begin{bmatrix} -150 + 300 - 30 \\ 2700 - 4800 + 180 \\ -8400 + 8400 - 240 \end{bmatrix} = -\frac{1}{240}\begin{bmatrix} 120 \\ -1920 \\ -240 \end{bmatrix}$$

10 Simplify. ✓ ½

$a = -\dfrac{1}{2}, \quad b = 8, \quad c = 1$

11 Hence the path of the ball is $y = -\frac{1}{2}x^2 + 8x + 1$. ✓ ½

Answer $a = -\frac{1}{2}$, $b = 8$, $c = 1$; **path:** $y = -\frac{1}{2}x^2 + 8x + 1$

Watch out: Writing (x, y) as (a, b); the unknowns are a, b, c and each point gives one equation.

16 **Long answer** CBSE SQP 2025-26 5 marks

For two matrices $A = \begin{bmatrix} 3 & -6 & -1 \\ 2 & -5 & -1 \\ -2 & 4 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & -2 & -1 \\ 0 & -1 & -1 \\ 2 & 0 & 3 \end{bmatrix}$, find the product AB and hence solve the system of equations: $3x - 6y - z = 3$, $2x - 5y - z + 2 = 0$, $-2x + 4y + z = 5$.

Formula / idea: If $AB = I$, then $A^{-1} = B$.

1 Row 1 of AB. ✓ ½
$(3 + 0 - 2,\ -6 + 6 + 0,\ -3 + 6 - 3) = (1,\ 0,\ 0)$

2 Row 2 of AB. ✓ ½
$(2 + 0 - 2,\ -4 + 5 + 0,\ -2 + 5 - 3) = (0,\ 1,\ 0)$

3 Row 3 of AB. ✓ ½
$(-2 + 0 + 2,\ 4 - 4 + 0,\ 2 - 4 + 3) = (0,\ 0,\ 1)$

4 So AB is the identity matrix. ✓ ½
$AB = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I$

5 Hence $A^{-1} = B$. ✓ 1
$AB = I \Rightarrow A^{-1} = B$

6 Write the system as $AX = C$; note $2x - 5y - z = -2$. ✓ ½
$\begin{bmatrix} 3 & -6 & -1 \\ 2 & -5 & -1 \\ -2 & 4 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 3 \\ -2 \\ 5 \end{bmatrix}$

7 $X = A^{-1}C = BC$. ✓ 1
$X = \begin{bmatrix} 1 & -2 & -1 \\ 0 & -1 & -1 \\ 2 & 0 & 3 \end{bmatrix} \begin{bmatrix} 3 \\ -2 \\ 5 \end{bmatrix} = \begin{bmatrix} 3 + 4 - 5 \\ 0 + 2 - 5 \\ 6 + 0 + 15 \end{bmatrix} = \begin{bmatrix} 2 \\ -3 \\ 21 \end{bmatrix}$

8 Hence $x = 2$, $y = -3$, $z = 21$. ✓ ½

Answer $AB = I$; $x = 2$, $y = -3$, $z = 21$

Watch out: Using $+2$ on the right of the second equation; move it across first: $2x - 5y - z = -2$.

17 **Long answer** CBSE 2023 5 marks

Solve the following system of equations by the matrix method: $x + 2y + 3z = 6$, $2x - y + z = 2$, $3x + 2y - 2z = 3$.

Formula / idea: $AX = B \Rightarrow X = A^{-1}B$, where $A^{-1} = \dfrac{1}{|A|}\operatorname{adj} A$.

1	Write as $AX = B$. $\begin{bmatrix}1 & 2 & 3\\ 2 & -1 & 1\\ 3 & 2 & -2\end{bmatrix}\begin{bmatrix}x\\ y\\ z\end{bmatrix} = \begin{bmatrix}6\\ 2\\ 3\end{bmatrix}$	✓ ½
2	Find $\lvert A\rvert$ by expanding along R_1. $\lvert A\rvert = 1(2-2) - 2(-4-3) + 3(4+3) = 0 + 14 + 21 = 35 \neq 0$	✓ 1
3	Cofactors of row 1. $A_{11} = 0, \quad A_{12} = 7, \quad A_{13} = 7$	✓ ½
4	Cofactors of row 2. $A_{21} = 10, \quad A_{22} = -11, \quad A_{23} = 4$	✓ ½
5	Cofactors of row 3. $A_{31} = 5, \quad A_{32} = 5, \quad A_{33} = -5$	✓ ½
6	$\operatorname{adj} A$ is the transpose of the cofactor matrix. $\operatorname{adj} A = \begin{bmatrix}0 & 10 & 5\\ 7 & -11 & 5\\ 7 & 4 & -5\end{bmatrix}$	✓ ½
7	Write A^{-1}. $A^{-1} = \dfrac{1}{35}\begin{bmatrix}0 & 10 & 5\\ 7 & -11 & 5\\ 7 & 4 & -5\end{bmatrix}$	✓ ½
8	Find $X = A^{-1}B$. $X = \dfrac{1}{35}\begin{bmatrix}0+20+15\\ 42-22+15\\ 42+8-15\end{bmatrix} = \dfrac{1}{35}\begin{bmatrix}35\\ 35\\ 35\end{bmatrix} = \begin{bmatrix}1\\ 1\\ 1\end{bmatrix}$	✓ ½
9	Hence $x = 1, y = 1, z = 1$.	✓ ½

Answer $x = 1, y = 1, z = 1$

Watch out: Forgetting to transpose the cofactor matrix when writing $\operatorname{adj} A$.

18 **Long answer** PYQ-style 5 marks

Gautam buys 5 pens, 3 bags and 1 instrument box and pays ₹160. From the same shop, Vikram buys 2 pens, 1 bag and 3 instrument boxes and pays ₹190. Ankur buys 1 pen, 2 bags and 4 instrument boxes and pays ₹250. Using the matrix method, find the cost of one pen, one bag and one instrument box.

Formula / idea: $AX = B \Rightarrow X = A^{-1}B$, where $A^{-1} = \dfrac{1}{|A|}\operatorname{adj} A$.

1	Let one pen, bag, box cost ₹x, ₹y, ₹z. $5x + 3y + z = 160, \quad 2x + y + 3z = 190, \quad x + 2y + 4z = 250$	✓ ½
2	Write as $AX = B$. $\begin{bmatrix}5 & 3 & 1\\ 2 & 1 & 3\\ 1 & 2 & 4\end{bmatrix}\begin{bmatrix}x\\ y\\ z\end{bmatrix} = \begin{bmatrix}160\\ 190\\ 250\end{bmatrix}$	✓ ½
3	Find $\lvert A\rvert$ by expanding along R_1. $\lvert A\rvert = 5(4-6) - 3(8-3) + 1(4-1) = -10 - 15 + 3 = -22 \neq 0$	✓ ½
4	Cofactors of row 1. $A_{11} = -2, \quad A_{12} = -5, \quad A_{13} = 3$	✓ ½
5	Cofactors of row 2. $A_{21} = -10, \quad A_{22} = 19, \quad A_{23} = -7$	✓ ½
6	Cofactors of row 3. $A_{31} = 8, \quad A_{32} = -13, \quad A_{33} = -1$	✓ ½
7	$\operatorname{adj} A$ is the transpose of the cofactor matrix. $\operatorname{adj} A = \begin{bmatrix}-2 & -10 & 8\\ -5 & 19 & -13\\ 3 & -7 & -1\end{bmatrix}$	✓ ½
8	Write A^{-1}. $A^{-1} = -\dfrac{1}{22}\begin{bmatrix}-2 & -10 & 8\\ -5 & 19 & -13\\ 3 & -7 & -1\end{bmatrix}$	✓ ½
9	Find $X = A^{-1}B$. $X = -\dfrac{1}{22}\begin{bmatrix}-320 - 1900 + 2000\\ -800 + 3610 - 3250\\ 480 - 1330 - 250\end{bmatrix} = -\dfrac{1}{22}\begin{bmatrix}-220\\ -440\\ -1100\end{bmatrix} = \begin{bmatrix}10\\ 20\\ 50\end{bmatrix}$	✓ ½
10	Hence a pen costs ₹10, a bag ₹20 and an instrument box ₹50.	✓ ½

Answer **Pen ₹10, bag ₹20, instrument box ₹50**

Watch out: Mixing up rows: each person's purchase is one equation, so it forms one row of A.

19 **Long answer** CBSE 2025 Compartment 5 marks

If $A = \begin{bmatrix} 3 & 2 & 1 \\ 4 & -1 & 2 \\ 7 & 3 & -3 \end{bmatrix}$, find A^{-1}. Using A^{-1}, solve the system of equations: $3x + 4y + 7z = 14$, $2x - y + 3z = 4$, $x + 2y - 3z = 0$.

Formula / idea: The coefficient matrix here is A', and $(A')^{-1} = (A^{-1})'$.

1 Find $|A|$ by expanding along R_1. ✓ ½

$|A| = 3(3-6) - 2(-12-14) + 1(12+7) = -9 + 52 + 19 = 62 \neq 0$

2 Cofactors of row 1. ✓ ½

$A_{11} = -3, \quad A_{12} = 26, \quad A_{13} = 19$

3 Cofactors of row 2. ✓ ½

$A_{21} = 9, \quad A_{22} = -16, \quad A_{23} = 5$

4 Cofactors of row 3. ✓ ½

$A_{31} = 5, \quad A_{32} = -2, \quad A_{33} = -11$

5 $\text{adj}\, A$ is the transpose of the cofactor matrix. ✓ ½

$\text{adj}\, A = \begin{bmatrix} -3 & 9 & 5 \\ 26 & -16 & -2 \\ 19 & 5 & -11 \end{bmatrix}$

6 Write A^{-1}. ✓ ½

$A^{-1} = \frac{1}{62} \begin{bmatrix} -3 & 9 & 5 \\ 26 & -16 & -2 \\ 19 & 5 & -11 \end{bmatrix}$

7 The columns of A are the rows of the system, so it reads $A'X = B$. ✓ ½

$A'X = B, \quad B = \begin{bmatrix} 14 \\ 4 \\ 0 \end{bmatrix}$

8 So $X = (A')^{-1}B = (A^{-1})'B$. ✓ ½

$X = \frac{1}{62} \begin{bmatrix} -3 & 26 & 19 \\ 9 & -16 & 5 \\ 5 & -2 & -11 \end{bmatrix} \begin{bmatrix} 14 \\ 4 \\ 0 \end{bmatrix}$

9 Multiply. ✓ ½

$X = \frac{1}{62} \begin{bmatrix} -42 + 104 + 0 \\ 126 - 64 + 0 \\ 70 - 8 + 0 \end{bmatrix} = \frac{1}{62} \begin{bmatrix} 62 \\ 62 \\ 62 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$

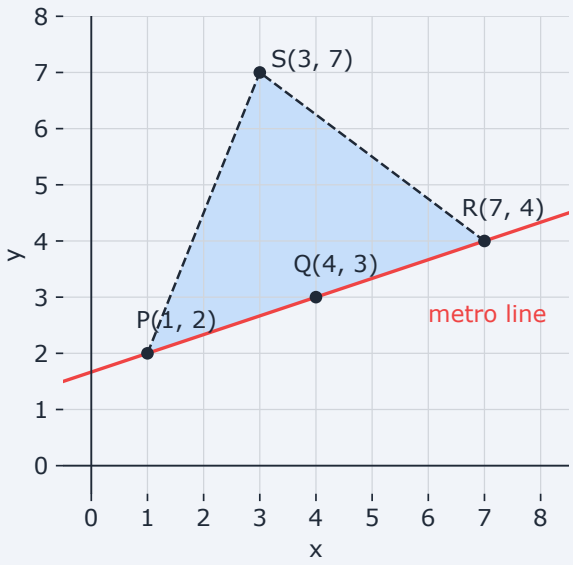
10 Hence $x = 1, y = 1, z = 1$. ✓ ½

Answer $A^{-1} = \frac{1}{62}\begin{bmatrix} -3 & 9 & 5 \\ 26 & -16 & -2 \\ 19 & 5 & -11 \end{bmatrix}$; $x = 1, y = 1, z = 1$

Watch out: Using $X = A^{-1}B$ directly; the coefficient matrix is A', not A.

20 **Case study** Board-pattern 4 marks

Case Study: A city planner marks three metro stations on a map grid at $P(1, 2)$, $Q(4, 3)$ and $R(7, 4)$. A new park is planned at $S(3, 7)$ (1 unit $= 100$ m).



(i) Using determinants, show that the stations P, Q and R lie on one straight line. 1 mark

(ii) Using determinants, find the area, in square units, of the triangular plot with corners P, R and S. 1 mark

(iii) Using determinants, find the equation of the line through P and R. 2 marks

Formula / idea: Area $= \frac{1}{2}\left|\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}\right|$; area 0 means collinear.

(i) 1 mark

1 Find the area determinant for P, Q, R. ✓ ½

$$\begin{vmatrix} 1 & 2 & 1 \\ 4 & 3 & 1 \\ 7 & 4 & 1 \end{vmatrix} = 1(3-4) - 2(4-7) + 1(16-21) = -1 + 6 - 5 = 0$$

2 Area is 0, so P, Q, R lie on one straight line. ✓ ½

Answer (i) **Determinant $= 0$, so P, Q, R are collinear.**

(ii) 1 mark

1 Write the area of $\triangle PRS$.

$$\Delta = \frac{1}{2}\begin{vmatrix} 1 & 2 & 1 \\ 7 & 4 & 1 \\ 3 & 7 & 1 \end{vmatrix} = \frac{1}{2}\left[1(4-7) - 2(7-3) + 1(49-12)\right]$$

✓ ½

2 Simplify.

$$\Delta = \frac{1}{2}(-3 - 8 + 37) = \frac{26}{2} = 13 \text{ sq units}$$

✓ ½

Answer (ii) 13 **sq units**

(iii) 2 marks

1 Let (x, y) be any point on PR. The area with P and R is 0.

$$\begin{vmatrix} x & y & 1 \\ 1 & 2 & 1 \\ 7 & 4 & 1 \end{vmatrix} = 0$$

✓ ½

2 Expand along R_1.

$$x(2-4) - y(1-7) + 1(4-14) = 0$$

✓ ½

3 Simplify.

$$-2x + 6y - 10 = 0$$

✓ ½

4 Divide by -2. Required line: $x - 3y + 5 = 0$.

✓ ½

Answer (iii) $x - 3y + 5 = 0$

Watch out: Leaving the area negative; area is the absolute value of the determinant expression.